



SAFETY DATA SHEET

according to regulation 1907/2006/EC (REACH) and 1272/2008/EC **MOL-LUB Ltd.**

Trade name: **MOL Essence 10W-40 semi-synthetic motor oil**

Version: 2 Latest revision: 29. 05.2015 Date of issue: 02. 04. 2013 Page: 1/(13)

1. Identification of the mixture and of the company/undertaking

1.1 Product identifier:

MOL Essence 10W-40 semi-synthetic motor oil

1.2 Relevant identified uses of the mixture and uses advised against

Relevant identified uses: semi-synthetic motor oil

Uses advised against: no data

1.3 Details of the supplier of the safety data sheet:

MOL-LUB Lubricant Production Distribution and Service Limited Liability Company

H-2931 Almásfüzitő, Fő út 21., Hungary

Phone / Fax: +36 34 526 330 / +36 34 526 391

E-mail: kenoanyag@mol.hu

Request SDS of:

MOL-LUB Lubricant Production Distribution and Service Limited Liability Company

Customer Service Center

H-2931 Almásfüzitő, Fő út 21., Hungary

Phone / Fax: +36 80 201 296 / +36 34 348 010

Responsible for SDS:

MOL-LUB Ltd. Csaba Horváth, head of SD, HSE & Business Support

Phone: +36 34 526 343; Mobile: +36 20 474 2644

e-mail: csahorvath@mol.hu

1.4 Emergency telephone number

Emergency telephone (07-15²⁰ h): +36 34 526 210 (CET) on workdays

Health Toxicological Information Service (ETTSZ 1096 Budapest, Nagyvárad tér 2.)

Tel.: +36 1 476 6464, or +36 80 201 199

National Health Toxicological Information Service:



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2. Hazards identification

2.1 Classification of the mixture

2.1.1 Classification of the mixture according to regulation 1272/2008/EC

Hazard Class and Category: Hazard statement:

-

2.1.2 Classification of the substance according to regulation 1999/45/EC

Human health hazards: Danger symbol not required.

Note: Prolonged and/or repeated contact may cause irritation on skin or in eyes depending on individual sensitivity, and/or skin sensibilization (See also Protective equipment).

Safety hazards: Danger symbol not required.

Environmental hazards: Danger symbol not required.

Note: Spills may form a film on water surfaces causing impaired oxygen transfer.

2.2 Label elements 1272/2008/EC

Product identification: Trade name: **MOL Essence 10W-40 semi-synthetic motor oil**

Hazardous components: -

GHS Pictogram: -

Signal word: -

Hazard statement:

Not required.

Supplemental hazard information:

EUH208 Contains Calcium long chain alkaryl sulphonate. May produce an allergic reaction.

Precautionary statements – Prevention:

P273 Avoid release to the environment.

Precautionary statements – Disposal:

P501 Dispose of contents/container in accordance with national regulation.

Other liabilities for labelling:

Tactile warning of danger: Not required.

Transport classification: see section 14.



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2.3 Other hazards

The product does not contain any PBT or vPvB substance according to annex XIII of regulation (EC) 1907/2006.

3. Composition/information on ingredients

3.2 Mixtures

Chemical description: Mixture of refined mineral base oils and synthetic oil containing additives.

Component(s) / Hazardous component(s):

Name	EU number	CAS number	Hazard symbol / Hazard classes and cat.	Risk phrase / Hazard statements	Conc. %(m/m)
Lubricating oils (petroleum), C24-C50* REACH Registr. Nr.: 01-2119489969-06	309-877-7	101316-72-7	67/548/EEC: - (Note L) 1272/2008/EC: Asp. Tox. 1 (Note L)	- H304	max. 70
Lubricating oils (petroleum), C18-40* REACH Registr.Nr.: 01-2119486987-11	305-594-8	94733-15-0	67/548/EEC: - (Note L) 1272/2008/EC: Asp.Tox.1 (Note L)	- H304	
Base oil – unspecified, hydrotreated heavy paraffinic, C20-C50*  REACH Registr. Nr.: 01-2119484627-25	265-157-1	64742-54-7	67/548/EEC: Xn (Note L) 1272/2008/EC: Asp.Tox.1 (Note L)	R 65 H304	max. 20
Base oil – unspecified, hydrotreated neutral oil-based, C20-C50 REACH Registr. Nr.: 01-2119474889-13	276-738-4	72623-87-1	67/548/EEC: - (Note L) 1272/2008/EC: Asp.Tox.1 (Note L)	- H304	
Mineral oil raffinate*		confidential	67/548/EEC: - (Note L) 1272/2008/EC: - (Note L)	- -	max. 10



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Name	EU number	CAS number	Hazard symbol / Hazard classes and cat.	Risk phrase / Hazard statements	Conc. %(m/m)
Polyolefin polyamine succinimide, polyol	polymer		67/548/EEC: - 1272/2008/EC: Aquatic Chronic 4	R 53 H413	max. 2.8
Zinc alkyl dithiophosphate  	272-028-3	68649-42-3	67/548/EEC: Xi, N 1272/2008/EC: Eye Dam. 1 Aquatic Chronic 2	R 41, R 51/53 H318 H411	< 1
Calcium long chain alkaryl sulphonate 	polymer	722503-686	67/548/EEC: Xi 1272/2008/EC: Skin Sens. 1 Aquatic Chronic 4	R 43, R 53 H317 H413	max. 0.75
Branched alkyl-carbolic acid and calcium alkyl-carbolic acid   	polymer	74499-35-7 and 132752-19-3	67/548/EEC: T, Xi, N 1272/2008/EC: Repr. 1B Eye Irrit. 20 Skin Irrit. 2 Aquatic Chronic 1	R 60 (Repr.cat. 2), R 36/38, R 50/53 H360 H319 H315 H410	max. 0.15

*: with exposure limit

The full text of each relevant R-, H- phrase and Hazard classes and cat. see in Section 16.

4. First aid measures

4.1 Description of first aid measures

General information: Never give anything by mouth to an unconscious person, or never induce vomiting.

Inhalation: Remove the affected person to fresh air. If rapid recovery does not occur, obtain medical attention.

Skin contact: Remove contaminated clothing. Wash skin with large amounts of water, use soap. In case of persistent irritation, get medical attention.



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Eye contact: Flush eyes with plenty of water for 10-15 minutes. In case of persistent irritation, get medical attention.

Ingestion: If swallowed, give water. **Do not** induce vomiting. Get medical attention.

Protection of first-aid person: No data available.

4.2 Most important symptoms and effects, both acute and delayed
No data available.

4.3 Indication of any immediate medical attention and special treatment need
Not required.

5. Fire-fighting measures

Fire hazards:
Combustible.

5.1 Extinguishing media
Suitable extinguishing media:
Foam, carbon dioxide, dry chemical powder.

Unsuitable extinguishing media:
Water jet.

5.2 Special hazards arising from the mixture
Hazardous combustion products:
On burning, carbon dioxide, carbon monoxide, sulphur oxides, phosphor oxides, various hydrocarbons and soot can be formed.

5.3 Advice for fire-fighters
Special protective equipment:
According to the existing fire-fighting regulations. Respiratory protection.

Further information:
Collect contaminated fire fighting water separately. It must not enter the sewage system.
Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures
Personal precautions: see Section 8.



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6.2 Environmental precautions:

Confine spills to prevent material from entering sewers, watercourses, drains and into soil. Notify relevant authority.

6.3 Methods and material for containment and cleaning up

On soil: All kind of ignition sources should be removed. Recover free liquid by pumping. Contain the rest or small quantities with sand, earth or other suitable absorbents. Dispose of according to local regulations.

On water: Confine the spillage. Remove from surface by skimming or suitable absorbents. Notify local authorities according to regulations.

6.4 Reference to other sections

Personal precautions: see section 8.

Waste treatment methods: see section 13.

7. Handling and storage

7.1 Precautions for safe handling

Keep general measures applied for normal operations with lubricants and flammable liquids.

Keep away from radiant heat and open flame.

Avoid contact with skin and eyes. Avoid prolonged breathing of oil vapours or mists.

Ensure washing facilities after working hours and before breaks. Take off contaminated or oil-soaked clothing, wash with warm water and soap.

When using do not eat, drink or smoke. Avoid splashing the product.

Handling temperature: not known

7.2 Conditions for safe storage, including any incompatibilities

Storage facilities must comply with regulations for storing of flammable liquids.

Store in dry, well ventilated place in original, closed containers.

Keep away from radiant heat, open flame and strong oxidizing agents.

Storage temperature: max. 40°C

7.3 Specific end use(s)

Semi-synthetic motor oil.

8. Exposure controls / personal protection

Engineering control measures:

Not required.



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8.1 Control parameters:

Mineral oil mist: TWA: **5 mg/m³**; STEL: 10 mg/m³, for oil mist, vapour excluded (ACGIH).

Method of testing, recommended: NIOSH 5026

8.2 Exposure controls

Personal protection:

Respiratory protection: Breathing apparatus not required.

Hand protection: Oil resistant gloves (EN 374, Breakthrough time 480 min) (e.g. nitrile rubber – minimal thickness 0.33 mm).

Note: Manufacturer's directions for use and the conditions of application should be observed.

Eye protection: Protective goggles not required.

Skin protection: Protective clothing (oil resistant).

Other special: no data

Environmental exposure controls:

Do not discharge into drains/surface waters/groundwater.

9. Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance:

Physical state:	liquid
Colour:	brown, clear
Odour:	characteristic

Change in physical state:

Pour point (ISO 3016):	typ. -33°C
Boiling point:	not available

Others:

Flash point (COC) (EN ISO 2592):	typ. 220°C
Ignition point (EN ISO 2592):	not available
Autoignition temperature:	not available
Explosive properties:	not explosive
Oxidizing properties:	not oxidize
Vapour pressure at 20°C:	negligible
Density at 15°C (EN ISO 12185):	0.860 – 0.880 g/cm ³
Solubility in water:	practically insoluble in water



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Solubility in other solvents:	gasoline, kerosene, toluene, etc.
n-Octanol/water partition coefficient:	not available
Vapour density:	not available
Heating value:	inf. 38 000 kJ/kg
Kinematic viscosity at 40°C (EN ISO 3104):	typ. 97.2 mm ² /s
Kinematic viscosity at 100°C (EN ISO 3104):	typ. 14.2 mm ² /s
pH:	not applicable

- 9.2 Other information
no data available

10. Stability and reactivity

- | | |
|--|---|
| 10.1 Reactivity: | Dangerous reactivity not known. |
| 10.2 Chemical stability: | No decomposition if stored and handled properly. |
| 10.3 Possibility of hazardous reactions: | Not known. |
| 10.4 Conditions to avoid: | Direct heat or ignition sources. |
| 10.5 Incompatible materials: | Strong oxidizing agents. |
| 10.6 Hazardous decomposition products: | No dangerous decomposition products are formed under normal conditions. Hazardous combustion products: See Section 5. |

11. Toxicological information

11.1 Information on toxicological effects

Acute toxicity:

Oral:	LD ₅₀ (rat)	> 2000	mg/kg	(based on components)
Dermal:	LD ₅₀ (rabbit)	> 2000	mg/kg	(based on components)

Acute toxicity: irritation

Skin:	not irritant (based on components)
Eye:	not irritant (based on components)
Note:	Prolonged and/or repeated contact may cause irritation on skin or in eyes depending on individual sensitivity.

Respiratory or skin sensitisation: not sensitising (based on components), but may produce an allergic reaction



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Other information, specific effects:

The product does not contain PCBs, PCTs, and other chlorine compounds, and heavy metals, barium compounds.

Note L: The classification as a carcinogen need not apply according to 1272/2008/EC, because it can be shown that the substance contains less than 3 % DMSO extract as measured by IP 346.

Germ cell mutagenicity:	not known, resp. not mutagen (based on components)
Carcinogenicity:	not known, resp. not carcinogen (based on components)
Reproductive toxicity:	not known, resp. no reproduction-damaging effect (based on components)
STOT-single exposure:	not known
STOT-repeated exposure:	not known
Aspiration hazard:	not known

12. Ecological information

12.1	Toxicity	No data for the product.
	Aquatic organisms:	
	Soil organisms:	
	Plants:	
12.2	Persistence and degradability	No data available.
	Biodegradability:	No data available.
12.3	Bioaccumulative potential	No data available.
12.4	Mobility	
	Mobility in soil	Absorbs in soil.
	Mobility in water:	Floats on water.
12.5	Results of PBT and vPvB assessment	Does not contain PBT and vPvB substances.
12.6	Other adverse effects	
	Heavy metal content:	None.
	PCT, PCB and other chlorinated hydrocarbons:	None.
	Environmental effects:	Spills may form a film on water surfaces causing impaired oxygen transfer.
	Water hazard class (German):	



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13. Disposal considerations

13.1 Waste treatment methods

Product disposal:

Wastes of the product or used oil should be treated as hazardous waste.

EWC cod: 13 02 05*

Mineral-based non-chlorinated engine, gear and lubricating oils.

EWC cod: 13 02 06*

Synthetic engine, gear and lubricating oils.

Disposal must be in compliance with national and local regulations.

Recommended waste treatment method: incineration

Packaging disposal:

Containers with product residue should also be treated as hazardous waste according to national and local disposal regulations.

EWC cod: 15 01 10*

Packaging containing residues of or contaminated by dangerous substances.

Disposal must be in compliance with national and local regulations.

Wastewater:

Quality of wastewater emitted to natural water must comply with national and local regulations.

Care should be taken in any case to ensure compliance with EC, national and local regulations. It is the responsibility of the user to know all relevant national and local regulations.

14. Transport information

Land transport:

Road/ Railway

ADR/RID:

Not classified.

14.1. UN number

14.2. UN proper shipping name

14.3. Transport hazard class(es)

14.4. Packing group

14.5. Environmental hazards

14.6. Special precautions for user

Waterways:

Inland waterways/ Sea transport ADN/IMDG:

Not apply to the product.

Air transport:

ICAO / IATA:

Not apply to the product.



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15. Regulatory information

- 15.1 Safety, health and environmental regulations/legislation specific for the mixture.
This safety data sheet has been prepared according to Regulation (EC) No 1907/2006 (mod.: 453/2010/EC) and to Regulation (EC) 1272/2008.
- 15.2 Chemical safety assessment.
not available

16. Other information

The information given in this data sheet is based on our best knowledge at the time of publication. The information is related only to this product and is intended to assist its safe transport, handling and use. The given physical and chemical parameters describe the product only for the purpose of safety requirements and therefore should not be construed as guaranteeing any specific property of the product or as being part of a product specification or any contract.

The manufacturer or supplier shall not take responsibility for any damages from the use other than recommended or other misuse of the product. It is the responsibility of the user to keep regulatory precautions and observe recommendations for safe use of the product.

Source of data presented in this material safety data sheet:

Test results of this product
Material safety data sheets of product's components
Hungarian and EU lists of dangerous substances
Relevant Hungarian regulation and EU directives

Classification for mixtures and used evaluation method according to regulation 1272/2008/EC (CLP)

- -

The full text of each relevant R-, H- phrase and Hazard classes and cat. in Section 3.:

R 36/38	Irritating to eyes and skin.
R 41	Risk of serious damage to eyes.
R 43	Bőrrel érintkezve túlérzékenységet okozhat (szenzibilizáló hatású lehet)
R 50/53	Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.



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R 51/53	Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
R 53	May cause long-term adverse effects in the aquatic environment.
R 60	May impair fertility.
R 65	Harmful: may cause lung damage if swallowed.
H304	May be fatal if swallowed and enters airways.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H360	May damage fertility or the unborn child.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H413	May cause long lasting harmful effects to aquatic life.
Asp.Tox.1	Aspiration hazard Category 1
Skin Irrit. 2	Skin corrosion/irritation Category 2
Skin Sens.1	Respiratory/skin sensitization Category 1
Eye Dam. 1	Serious eye damage/eye irritation Category 1
Eye Irrit. 2	Causes serious eye irritation.
Repr. 1B	Reproductive toxicity Category 1B
Aquatic Chronic 1	Hazardous to the aquatic environment, Chronic Category 1
Aquatic Chronic 2	Hazardous to the aquatic environment, Chronic Category 2
Aquatic Chronic 4	Hazardous to the aquatic environment, Chronic Category 4

Legend:

ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute Toxicity Estimate
BCF	Bioconcentration Factor
BOD	Biological Oxygen Demand
Bw	Body Weight
C&L	Classification and Labeling
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging (1272/2008/EC)
CMR	Carcinogenic, Mutagenic or toxic to Reproduction
COD	Chemical Oxygen Demand
CSA	Chemical Safety Assessment



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CSR	Chemical Safety Report
DMEL	Derived Minimal Effect Level
DNEL	Derived No Effect Level
ECHA	European Chemicals Agency
Ec _x	Effective Concentration x%
ErC ₅₀	EC ₅₀ in terms of reduction of growth rate
Ed _x	Effective Dose x%
EC	European Community
EC number	European Community number
ELINCS	European List of Notified Chemical Substances
ES	Exposure Scenario
ESIS	European Chemical Substances Information System
IARC	International Agency for Research on Cancer
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
LC _x	Lethal Concentration x%
LD _x	Lethal Dose x%) Halálós dózis x%
LOAEC	Lowest Observed Adverse Effect Concentration
LOAEL	Lowest Observed Adverse Effect Level
LOEC	Lowest Observed Effect Concentration
LOEL	Lowest Observed Effect Level
NOEC	No observed effect concentration
NOEL	No observed effect level
NLP	No-Longer Polymer
NOAEL	No Observed Adverse Effect Level
OECD	Organisation for Economic Cooperation and Development
PBT	Persistent Bioaccumulative and Toxic
PNEC	Predicted No-Effect Concentration
ppm	parts/million
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
RID	Regulations concerning the International carriage of Dangerous Goods by Rail
SVHC	Substance of Very High Concern
UVCB	substance of unknown or variable composition, complex reaction products or biological materials
VOC	Volatile organic compounds
vPvB	Very Persistent and very Bio-accumulative

Revision Indicators:

Section	Subject of change	Date	Version
1	Details of the supplier of the safety data sheet, Responsible for SDS	29.05.2015	2
2	Other hazards		
5	Fire hazards		
9	Physical and chemical properties		
14	Transport information		
1-16	Other corrections		